

Instacart Customer & Market Basket Analysis

Final Report

Role: Data Analyst · **Dataset:** Instacart Online Grocery Shopping Dataset 2017

Scope: 206,209 customers · 3,421,083 orders · 33,819,106 purchased items · 49,688 products

1. Executive Summary

Instacart's transaction data describes a **replenishment business, not a discovery business**. **59.0% of every item sold is something that customer has bought before**, and that single figure determines where the commercial opportunity sits.

Six findings carry the analysis:

1. **The repeat habit is built between a customer's 2nd and 6th order.** Reorder rate climbs from 27.2% to 54.1% across that window and then flattens. Retention effort spent after order 10 is spent on customers who are already retained.
2. **Order frequency and basket size are statistically unrelated** (Spearman $\rho = 0.06$). They are two independent growth levers, and treating "engagement" as one number conflates them.
3. **Customers run on a weekly clock.** 50.9% of repeat orders arrive within 7 days, with a distinct spike exactly at day 7 — a non-arbitrary trigger point that the business did not have to invent.
4. **Produce drives traffic; dairy drives the return visit.** Produce is 29.2% of units and reaches 74.9% of baskets, but dairy & eggs has the highest reorder rate of any department (67.0%).
5. **Reorder rate is a consumption-cadence meter, not a satisfaction score.** Category explains 56.3% of the variance in a product's reorder rate. Olive oil's 47.7% is healthy; a milk at 47.7% would be alarming.
6. **The best cross-sell signals are culinary, and the best-selling product is useless as a recommendation.** Garlic↔onion (lift 5.64) and pasta↔pasta sauce (lift 4.41) are real; bananas appear in 14.7% of baskets and therefore predict nothing — yet they are the consequent of 21.5% of all mined rules.

Six recommendations follow in §11, each tied to the evidence above. The highest-value one is a **replenishment subscription seeded from the milk and yogurt cluster**, where the behaviour already exists at an 84–86% reorder rate and the product is only being asked to stop making the customer retype it.

What this dataset cannot tell us: there are no prices, margins, dates or demographics. No revenue figure appears anywhere in this report, and seasonality cannot be measured.

2. Business Problem

How do Instacart customers behave, which products matter most to them, and which products should be marketed or recommended together?

Broken into the questions the analysis had to answer:

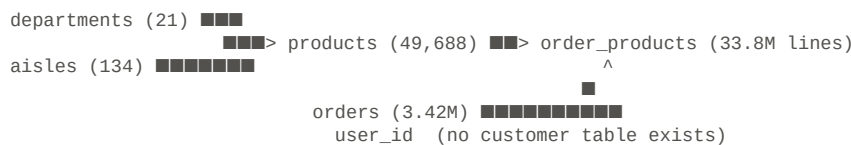
#	Question	Where answered
1	Who are the customers and how do they shop?	§6
2	Which products and categories matter most?	§7
3	When do customers buy?	§5
4	What drives repeat purchase?	§8
5	Which products are bought together?	§9

#	Question	Where answered
6	What commercial opportunities follow?	\$10–11

3. Dataset

Six tables, in a star-shaped schema with one large fact table split across two files.

Table	Rows	Grain	Role
orders	3,421,083	one order	fact header — customer, sequence, day, hour, interval
order_products__prior	32,434,489	product × order	basket contents, full history
order_products__train	1,384,617	product × order	basket contents, most recent order of 131,209 customers
products	49,688	one product	dimension; carries both aisle and department keys
aisles	134	one aisle	dimension
departments	21	one department	dimension



Two structural facts shaped the whole project.

There is no customer table. Every customer attribute in this report — order count, average basket, reorder rate, interval, segment — was derived from order history. Demographic segmentation was never an option.

75,000 orders have no contents. The test orders were withheld in the original Kaggle release. They are excluded from all item-level analysis and retained for order-timing analysis, where their fields are valid. Every table in this report names its base:

- **all orders (3,421,083)** — timing, frequency, interval
- **orders with contents (3,346,083)** — products, categories, baskets, reorder rate

4. Data Preparation

4.1 Data quality

The dataset is unusually clean: **zero orphan foreign keys, zero duplicate rows, zero invalid codes**, and the composite key (`order_id`, `product_id`) is unique in both basket files. **No row was deleted and no value was imputed in this entire project.**

What it needed was interpretation, not repair. Four fields are easy to misread:

#	Issue	Decision	Reasoning
1	206,209 nulls in <code>days_since_prior_order</code>	kept as null + <code>is_first_order</code> flag	Every null sits on <code>order_number</code> = 1 and the count equals the customer count exactly. The null means "no previous order exists". Imputing 0 would invent 206,209 same-day repeat purchases.

#	Issue	Decision	Reasoning
2	days_since_prior_order capped at 30 (10.8% of orders)	flagged dspo_is_capped; interval reported both ways	369,323 orders sit at exactly 30 against 18,418 at 29 — a 20× cliff no behaviour produces. "30" means "at least 30".
3	99 product names shared by multiple product_ids	added product_uid merging same-name-same-aisle ids	'BBQ Sauce' and 'Bbq Sauce' are one shelf item listed twice. Split, it ranked #12,002; merged, #9,842. Only 0.23% of the catalogue is affected, and product_id was left untouched.
4	missing / other placeholder categories (1,258 products)	flagged is_placeholder; kept in totals, excluded from rankings	An "unknown" bucket must never win a top-category ranking.

Also carried forward: order_number is capped at 100 (1,374 customers sit on it), and order_dow has **no documented day mapping** — day names in this report are a stated assumption, and no conclusion depends on them.

Outliers were examined and kept. The largest basket is 145 items and the busiest customer has 100 orders. Both are plausible grocery behaviour, and removing them would delete exactly the high-value behaviour the project exists to explain.

4.2 Integration

The required chain — Customer → Order → Order Product → Product → Aisle → Department — was built with one governing rule: **aggregate before joining**. Joining 33.8M basket lines onto 3.42M orders would fan the order table out to 33.8M rows and silently weight every order-level average by basket size. Basket size was therefore aggregated to order grain first, then joined back.

Every join changed the row count by exactly zero, which is the test that a many-to-one join on a primary key is correct. A left join (not inner) preserved the 75,000 test orders with a null basket size rather than dropping them.

5. Exploratory Analysis — When Customers Buy

Hour. 64.9% of all orders are placed between 09:00 and 16:00, peaking at 10:00 (288,418 orders). Overnight (00:00–05:00) is 1.8%. But **the busiest hour is not the biggest-basket hour**: 21:00–23:00 carries only 5.3% of orders and the largest baskets of the day (10.7–11.0 items vs 10.1 overall).

Day. Days 0 and 1 carry 34.7% of the week (600,905 + 587,478 orders); the quietest day carries 12.5%. Peak-to-trough is only 1.41×. Critically, **the two peak days have different shapes** — day 0 peaks 13:00–15:00 with 11.2-item baskets, day 1 peaks 09:00–11:00 with 10.2. A single weekend campaign sent at one time misses one of them.

Interval. 50.9% of repeat orders arrive within 7 days, and day 7 is a visible spike (10.0% of repeat orders, against 7.5% at day 6 and 5.7% at day 8), with secondary bumps at 14, 21 and 30. **Customers run a weekly shop.**

The interval also predicts basket composition: reorder rate is **65.8%** for orders placed within 3 days and **48.0%** after 25+ days. A customer returning after a week wants their usual list; one returning after a month is rebuilding the habit.

Cart position. Reorder rate falls from **67.9% at cart position 1** to 51.0% at position

- Customers add their habits first and browse later — which is a direct instruction about where in a session discovery belongs.

6. Customer Analysis

Frequency. Median 10 orders per customer, mean 16.6 (skew 2.4). Concentration is real but moderate: the top 10% of customers buy **35.1%** of items, the top 20% buy **53.7%**. This is *not* an 80/20 business — the bottom half still accounts for 17.8% of volume.

The key negative result. Order frequency and basket size are **uncorrelated** ($\rho = 0.06$). What *does* move with order count is reorder rate ($\rho = 0.73$) and, negatively, interval ($\rho = -0.60$). Across order bands, basket size moves only 9.6 → 10.4 items while reorder rate moves **23.0% → 73.7%** and interval falls **20.3 → 5.1 days**.

***Loyalty here means frequency and repetition, not basket size.** A "spend \$10 more" promotion pushes the lever that does not move.*

Segmentation. K-Means on the three dimensions the brief specifies (order count — log transformed because it is skewed 2.4 — average basket, reorder rate).

Stated honestly: the silhouette score peaks at $k = 2$ (0.365) and never exceeds 0.37, and the elbow curve has no sharp break. **This customer base is a continuum, not a set of natural tribes.** $k = 4$ was chosen because it separates the two independent levers into segments that can be acted on differently — a management partition, not a discovery.

Segment	Customers	% of items	Orders	Basket	Reorder	Interval
Loyal Regulars	44,475 (21.6%)	53.4%	40.6	10.1	69.0%	8.8 d
Big-Basket Stockers	33,020 (16.0%)	20.6%	10.6	19.4	46.2%	16.9 d
Steady Small-Basket	62,996 (30.5%)	17.1%	12.9	7.0	49.3%	15.6 d
Occasional / At-Risk	65,718 (31.9%)	8.9%	5.7	8.0	22.3%	19.1 d

One segment in five buys over half the volume. A third of the customer base produces a twelfth of it. The two middle segments have nearly the same order count and a 2.8× different basket — they need opposite interventions.

7. Product & Category Analysis

Top products. Bananas lead with 491,291 units bought by 76,125 customers (**36.9% of all customers**), then organic bananas (394,930). Nine of the top ten are fresh produce. The volume leaderboard and the reorder leaderboard are **nearly the same list** — these products are big because the same people buy them repeatedly, not because many people try them once.

Highest reorder rate (minimum 2,000 purchases, because a rate on 12 purchases is noise): nine of the top ten are **milk variants**, from 86.1% down to 83.9%, with bananas the only non-dairy entry. These are products people *run out of*, not products they *choose*.

Concentration. The top 100 products carry 23.1% of units and the top 1% carry 42.7% — but reaching 80% of volume still takes **4,548 products**. This is neither a hero-SKU business nor a flat long tail.

Volume vs loyalty. Splitting the 8,563 products with ≥500 purchases at the medians of that pool (1,287 units, 54.6% reorder rate):

Quadrant	Products	Units	Share	Example	Strategy
Core staples	2,537	20.4M	67.5%	Banana, baby spinach, whole milk	Never out of stock
Traffic drivers	1,745	6.4M	21.1%	Olive oil, green onions, ginger	Acquisition, basket fillers
Loyal niche	1,746	1.4M	4.7%	Specific cereals, granola bars	Personalised recommendation
Long tail	2,535	2.0M	6.6%	Paper plates, speciality cheese	Range completeness

Categories. Produce is 29.2% of units and reaches **74.9% of baskets**; dairy & eggs is 16.7% of units but has the **highest reorder rate of any department, 67.0%**. Together they are 45.9% of everything sold. At the bottom, personal care (32.2%) and pantry (34.7%) reorder least.

Most habitual aisle: **milk, 78.2%**. Least: **spices & seasonings, 15.3%**, followed by food storage (25.5%), cleaning products (29.0%) and baking ingredients (30.5%).

*Every low-reorder category on that list is a **slow-consumption** category. A jar of cumin lasts a year. This is cadence, not dissatisfaction.*

8. Reorder Analysis

59.0% of all basket lines are reorders; **62.9%** once first orders — which cannot contain a reorder by definition — are excluded.

The habit curve. Reorder rate by the customer's nth order:

Order #	1	2	3	4	5	6	10	20	50
Reorder rate	0%	27.2%	38.6%	45.4%	50.3%	54.1%	63.4%	73.6%	81.9%

The curve gains **27 percentage points between orders 2 and 6** and then flattens. **That is the habit-formation window**, and it is the most actionable timing finding in the project.

By customer tenure the spread runs from 24.0% (3–4 lifetime orders) to **75.2%** (51–100). Heavy customers also explore *more* in absolute terms — 169.8 distinct products versus 26.5 — so repetition and breadth grow together rather than trading off.

Reorder rate must be benchmarked within its category. A variance decomposition shows **aisle explains 56.3%** of the variance in a product's reorder rate and department 40.5%. The remaining ~44% is where genuine product-level preference lives. An olive oil at 47.7% is performing well for pantry (mean 32.9%); a milk at 60% is performing badly for dairy (mean 61.9%).

9. Market Basket Analysis

Method. FP-Growth (a prefix tree rather than Apriori's repeated scans) on a fixed random sample of **600,000 baskets** (seed 42), at two levels: the **top 300 products** (35.6% of units; 65.4% of baskets hold ≥ 2 of them) and **all 134 aisles**. Baskets containing none of the scoped items stay in the denominator, so support is never inflated by shrinking the universe.

Thresholds, justified from the data. Minimum support 0.001 (≥ 600 of the 600,000 baskets), minimum confidence 0.05, minimum lift 1.0. At support 0.005 only 95 rules survive and all are banana pairs; at 0.0005 the output is too rare to merchandise against. Confidence is deliberately low — with a median basket of 8 items drawn from 49,573 products, a 5% conditional probability is a strong signal, and the textbook 50% would return only tautologies. **Lift does the ranking.** Result: **3,048 product rules** and **3,692 aisle rules** (the brief asks for a minimum of 10).

The required rule table

Antecedent	Consequent	Support	Conf.	Lift
Total 2% Greek Yogurt Blueberry	Total 2% Strawberry + Total 2% Peach	0.12%	18.6%	75.58
Icelandic Skyr Blueberry	Non Fat Raspberry Yogurt	0.22%	38.0%	75.61
Total 2% Strawberry + Peach	Total 2% Blueberry	0.12%	61.6%	68.12

Antecedent	Consequent	Support	Conf.	Lift
Sparkling Lemon + Grapefruit Water	Lime Sparkling Water	0.14%	47.0%	32.36
Organic Garlic	Organic Yellow Onion	—	—	5.64 (pairwise)
dry pasta	pasta sauce	1.95%	27.4%	4.41
pasta sauce	dry pasta	1.95%	31.4%	4.41
Limes	Large Lemon	—	—	4.08 (pairwise)
canned meals/beans	canned jarred vegetables	1.83%	26.3%	3.55
Organic Cucumber	Organic Grape Tomatoes	—	—	3.77 (pairwise)
fresh herbs	fresh vegetables	7.96%	84.6%	1.90
Organic Hass Avocado	Bag of Organic Bananas	1.94%	29.1%	2.45

Business interpretation — three distinct patterns

Pattern 1 — Flavour variety within one brand (lift 30–76). The strongest associations in the dataset are between *different flavours of the same yogurt line* (lift up to 75.6, confidence up to 61.6%), with sparkling water showing the same shape. **These are not cross-sell opportunities** — the customer has already chosen the brand and is assembling a mixed pack inside a decision already made. They are worth something different: *multipack bundles* (converting four line items into one pick), *range protection* (delisting one flavour will damage the others), and *substitution logic* (if blueberry is out, the right substitute is strawberry from the same line, not another brand's blueberry).

Pattern 2 — Recipe complements (lift 3–6). These are the genuine cross-sell rules, found among products that are each individually popular, so rarity cannot explain the association: garlic↔onion **5.64**, limes↔lemon **4.08**, cucumber↔grape tomatoes **3.77**, lemon↔avocado 3.62. The customer's intent — *cook a meal* — is only partly expressed by the first item, which is exactly what makes a prompt useful rather than intrusive.

Pattern 3 — Category adjacency (high support). Aisle rules are far more robust, holding in 1–2% of *all* baskets: **dry pasta ↔ pasta sauce (lift 4.41)**, canned meals ↔ canned vegetables (3.55), and **fresh herbs → fresh vegetables at 84.6% confidence** — the highest single-antecedent confidence found at any level. These identify **shopping missions** rather than product affinities, which makes them the better basis for merchandising, campaign timing and warehouse pick-path design.

The negative finding matters commercially. Bananas are in 14.7% of baskets, so they co-occur with everything and predict nothing — yet they are the consequent of **21.5% of all mined rules**. A recommender ranked on confidence would recommend almost nothing else.

Limitations

1. Association is not causation — garlic and onion co-occur because both are in the recipe.
2. Support is relative to the 600,000-basket sample and the top-300 scope; the *ranking* is stable, the absolute support figures are scope-dependent.
3. **No price or margin data exists**, so every bundle recommendation is conditional on a margin check.
4. No dates, so seasonality is invisible and would be averaged into the annual figure.

10. Key Findings

Each stated as **Finding** → **Evidence** → **Business meaning**.

Finding 1 — This is a replenishment business. *Evidence:* 59.0% of all 33.8M items sold are reorders (62.9% excluding first orders), rising to 81.9% by a customer's 50th order. *Business meaning:* the core product problem is making the known list effortless, not surfacing novelty. Discovery features compete with the task

most customers came to do.

Finding 2 — The habit forms between orders 2 and 6. *Evidence:* reorder rate climbs 27.2% → 54.1% across that window, then flattens (63.4% at order 10, 81.9% at order 50). By tenure: 24.0% for 3–4 order customers vs 75.2% for 51–100. *Business meaning:* this is where retention spend has leverage. A customer who reaches their sixth order is behaving like a retained customer; one who stalls at three is not. After order 10 the spend is largely redundant.

Finding 3 — Frequency and basket size are independent levers. *Evidence:* Spearman $\rho = 0.06$ between order count and average basket; across order bands the basket moves 9.6 → 10.4 items while reorder rate moves 23.0% → 73.7%. *Business meaning:* a single "engagement score" conflates two unrelated behaviours. Steady Small-Basket customers (7.0 items, 12.9 orders) need basket building; Big-Basket Stockers (19.4 items, 10.6 orders) need frequency. One campaign cannot do both.

Finding 4 — Customers run on a weekly clock. *Evidence:* 50.9% of repeat orders within 7 days; a spike at exactly day 7 (10.0% vs 7.5% at day 6, 5.7% at day 8); secondary bumps at 14, 21, 30. *Business meaning:* the reminder trigger does not have to be invented or A/B-guessed — the customer base has already chosen it. Deviation from a customer's own established interval is a better churn signal than any fixed threshold.

Finding 5 — Produce drives traffic; dairy drives return. *Evidence:* produce = 29.2% of units and 74.9% basket penetration, reorder rate 65.1%; dairy & eggs = 16.7% of units but the highest reorder rate of any department, 67.0%, led by milk at 78.2%. *Business meaning:* two different roles justifying two different investments. Produce quality and availability drive acquisition and basket reach; dairy reliability drives frequency.

Finding 6 — Reorder rate measures consumption cadence, not satisfaction. *Evidence:* aisle explains 56.3% of the variance in product reorder rate (department 40.5%). The lowest-reordering aisles — spices 15.3%, food storage 25.5%, baking 30.5% — are all slow-consumption categories. *Business meaning:* never score a product against the overall 59%. Benchmark within its category, or every pantry item looks like a failure and every milk looks like a success.

Finding 7 — The real cross-sell signal is culinary, and the top seller is worthless for it. *Evidence:* garlic↔onion lift 5.64, pasta↔pasta sauce 4.41, lime↔lemon 4.08; meanwhile bananas are in 14.7% of baskets and are the consequent of 21.5% of all rules. *Business meaning:* recipe-shaped bundles and basket-completion prompts are evidence-backed. A confidence-ranked recommender is not — it would recommend bananas to everyone.

Finding 8 — Small orders are a disproportionate operational load. *Evidence:* 17.1% of orders contain 3 items or fewer but carry only 3.5% of units; 4.9% are a single item. These tiny orders have the *highest* reorder rate (65.0%) — they are forgotten-essentials top-ups. *Business meaning:* a minimum-order rule would suppress a habit signal. Basket building — prompting the customer's own usual items at checkout — addresses the economics without punishing the behaviour.

11. Business Recommendations

Every recommendation names the evidence it rests on and how it would be measured.

R1 — Launch replenishment subscriptions, seeded from the milk & yogurt cluster

Evidence. The nine highest reorder rates in the catalogue ($\geq 2,000$ purchases) are all milk variants, 83.9–86.1%. Milk is the most habitual aisle at 78.2%; dairy & eggs is the most habitual department at 67.0%. 50.9% of repeat orders arrive within 7 days. **Action.** Offer a weekly or fortnightly auto-add for a customer's top 3–5 repeat products, defaulting to their own observed interval rather than a fixed schedule, with one-tap skip. **Why it works.** The behaviour already exists at an 84%+ rate; the product is only removing the retyping. It does not ask the customer to change anything. **Measure.** Share of orders containing a subscribed item; interval variance before vs after; churn of subscribers vs a matched control. **Risk.** Over-delivery damages trust fast.

Default to skip-friendly and never auto-add a slow-consumption category (anything below ~40% reorder rate — pantry, spices, cleaning).

R2 — Concentrate retention spend on orders 2–6

Evidence. Reorder rate climbs 27.2% → 54.1% between orders 2 and 6 and then flattens. Occasional / At-Risk customers (31.9% of the base, 8.9% of items) average 5.7 orders and a 22.3% reorder rate — they stall exactly in this window. **Action.** A fixed onboarding sequence over a new customer's first six orders: reorder prompts from their own first-basket items, a reminder timed to their emerging interval, and incentives front-loaded to orders 2–4 rather than spread evenly. **Why it works.** It targets the only part of the lifecycle where the curve is steep. Spend after order 10 is largely spent on customers who were going to stay.

Measure. Share of new customers reaching order 6 within a fixed window; reorder rate at order 6 vs control.

Risk. The dataset has no dates, so "within 90 days" cannot be validated here — the window needs calibrating against live data before launch.

R3 — Treat the two middle segments as two different problems

Evidence. Order frequency and basket size are uncorrelated ($\rho = 0.06$). Steady Small-Basket: 62,996 customers, 7.0 items, 12.9 orders, 15.6-day interval. Big-Basket Stockers: 33,020 customers, 19.4 items, 10.6 orders, 16.9-day interval. **Action.** For Small-Basket customers, basket building — complete-your-basket prompts drawn from the recipe pairs in R4, and a reminder of their own usual items at checkout. For Stockers, frequency — a replenishment reminder at ~14 days, not a bigger cart. **Why it works.** Each pushes the lever that is actually movable for that group. A single "engagement" campaign would push the wrong one for half of them. **Measure.** Basket size for Small-Basket; order interval for Stockers. Each segment is judged only on its own metric. **Risk.** Segments are a partition of a continuum (silhouette ≤ 0.37), so boundaries will shift on re-run. Assign on current behaviour, re-score regularly, and never expose segment names to customers.

R4 — Build basket-completion prompts on recipe pairs, not on popularity

Evidence. Among best-selling products: garlic↔onion lift 5.64, lime↔lemon 4.08, cucumber↔grape tomatoes 3.77, lemon↔avocado 3.62. At aisle level: pasta↔pasta sauce lift 4.41 in 1.95% of baskets. Against this: bananas are the consequent of 21.5% of all rules while adding no information. **Action.** Ship three evidenced kits — aromatics (garlic, onion), salad (cucumber, tomatoes, lemon), pasta night (dry pasta, sauce, parmesan) — and a completion prompt that fires on the *gap*: garlic in the basket without onion. Explicitly exclude the top ~20 products by basket penetration from recommendation eligibility. **Why it works.** It targets intent the first item only partly expresses, which is why the prompt is useful rather than intrusive. Excluding bananas removes the single largest source of worthless recommendations. **Measure.** Prompt acceptance rate, and incremental units of the *consequent* — not total basket size, which drifts for unrelated reasons. **Risk. No margin data exists in this dataset.** Every bundle must pass a margin check before launch; a high-lift pair of two low-margin items is commercially worthless.

R5 — Set service levels by quadrant, and benchmark reorder rate within category

Evidence. 2,537 core staples (5% of purchased products) carry 67.5% of units in the ≥ 500 -purchase pool at a 64.8% average reorder rate. Aisle explains 56.3% of the variance in product reorder rate. Spices reorder at 15.3%, milk at 78.2% — and both are healthy. **Action.** Tier availability by quadrant: core staples get the highest service-level guarantee and substitution logic; traffic drivers are managed for breadth; long tail is reviewed for delisting *only* with a cannibalisation check. Replace the single company-wide reorder benchmark with a per-aisle one. **Why it works.** An availability failure on a core staple damages a repeat relationship, not just one sale. And a single benchmark currently flags every slow-consumption category as failing when it is behaving normally. **Measure.** In-stock rate weighted by quadrant; reorder rate tracked as a *deviation from aisle mean* rather than an absolute. **Risk.** Flavour-variety products (R4, Pattern 1) are bought as sets — the per-SKU profitability review that drives delisting will miss that removing one flavour damages the others.

R6 — Time campaigns to the two differently-shaped weekend peaks

Evidence. Days 0 and 1 carry 34.7% of the week, but day 0 peaks 13:00–15:00 with 11.2-item baskets while day 1 peaks 09:00–11:00 with 10.2. 64.9% of orders fall in 09:00–16:00. Evening orders (21:00–23:00) are only 5.3% of volume but carry the largest baskets (up to 11.0 items). **Action.** Two separate weekend sends timed to each peak's own shape — a stock-up message before the day-0 afternoon, a top-up message before the day-1 morning. Schedule capacity to the hour, not the day: peak-to-trough across days is only 1.41×, across hours it is 53×. **Why it works.** A single weekend campaign fires at the wrong time for one of two distinct shopping occasions. **Measure.** Open-to-order conversion by send window; basket size by window. **Risk.** Day names are an assumption — the mapping is undocumented in the source. Validate against live timestamps before committing spend to a named day.

12. Conclusion

The brief's closing questions, answered.

Who are Instacart's customers and how do they shop? 206,209 customers placing a median of 10 orders of a median 8 items, on a weekly rhythm concentrated in daytime hours. They divide into a loyal fifth that buys over half the volume, a frequent-but-light group, a large-basket-but-infrequent group, and a third of the base that never forms the habit.

Which products and categories matter most? Produce by reach (29.2% of units, 74.9% of baskets); dairy by loyalty (67.0% reorder rate). 2,537 core staples carry 67.5% of volume in the reliable pool. Bananas are the single most important product by every volume measure — and the least useful one to recommend.

When do customers buy? 09:00–16:00 (64.9% of orders), concentrated on two weekend days of different shapes, on a 7-day repeat clock.

What is the reorder pattern? 59.0% of all items. It is built between orders 2 and 6, compounds to 81.9% by order 50, and is governed far more by how fast a category is consumed than by how much a product is liked.

Which products are bought together? Three different things wearing one name: flavour variants of a brand (lift up to 75.6 — bundle these), recipe complements (lift 3–6 — cross-sell these), and category adjacencies (pasta↔sauce, herbs→vegetables — merchandise around these).

What opportunities follow, and what should the company do? Six evidence-backed recommendations in §11, led by replenishment subscriptions seeded from a cluster that already reorders at 84%+, and by concentrating retention spend on the orders 2–6 window where the habit curve is actually steep.

The honest caveat. This dataset contains no prices, no margins, no dates and no demographics. Nothing here can be converted into a revenue forecast, nothing can be checked for seasonality, and every bundle recommendation is conditional on a margin check this data cannot perform. What it can support — behavioural findings, timing, category structure and product association — is stated above, and nothing beyond that has been claimed.

Analysis by notebooks 01–06 · dashboard in dashboard/instacart_dashboard.xlsx · all figures in reports/figures/ · all supporting tables in reports/tables/.